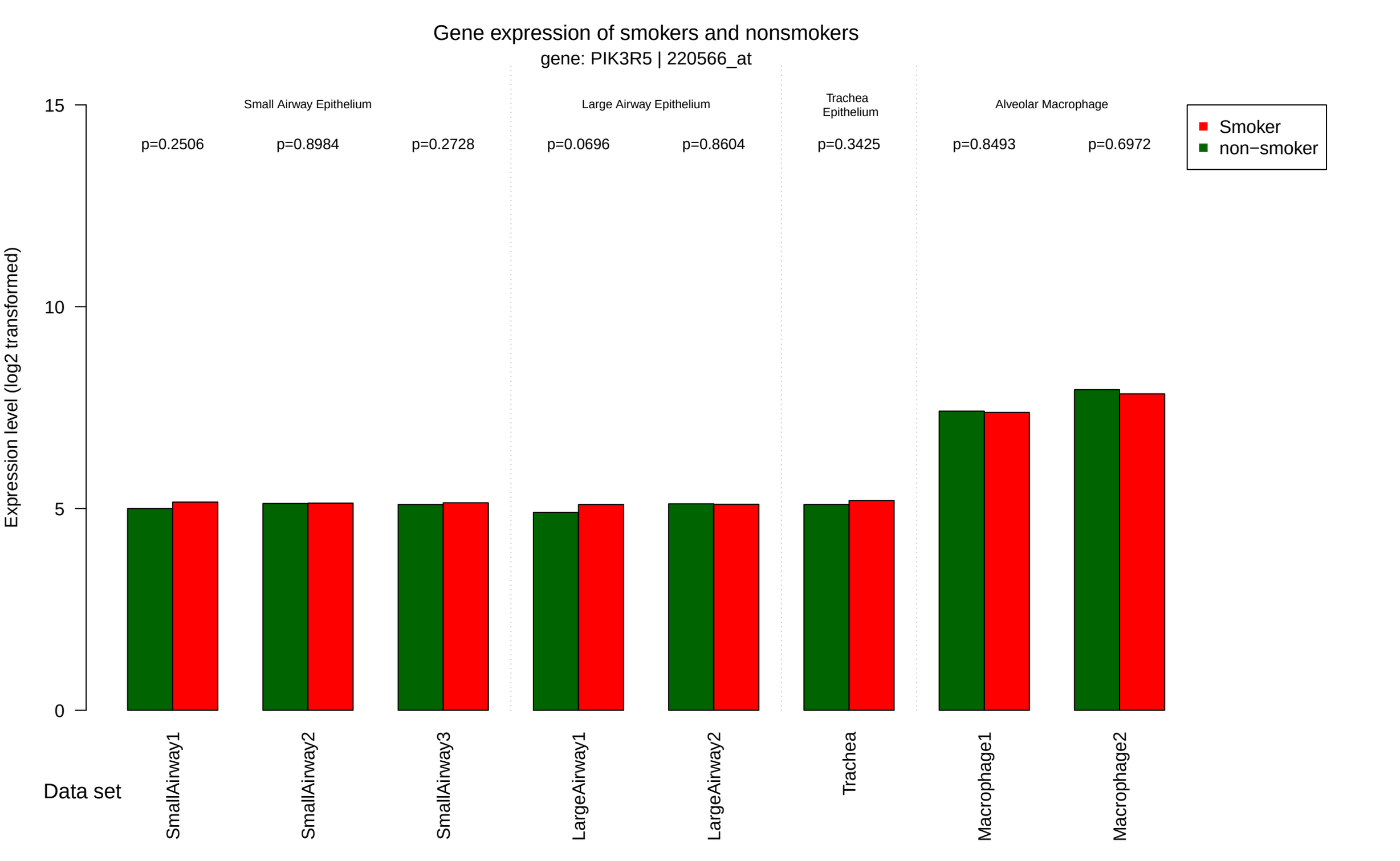
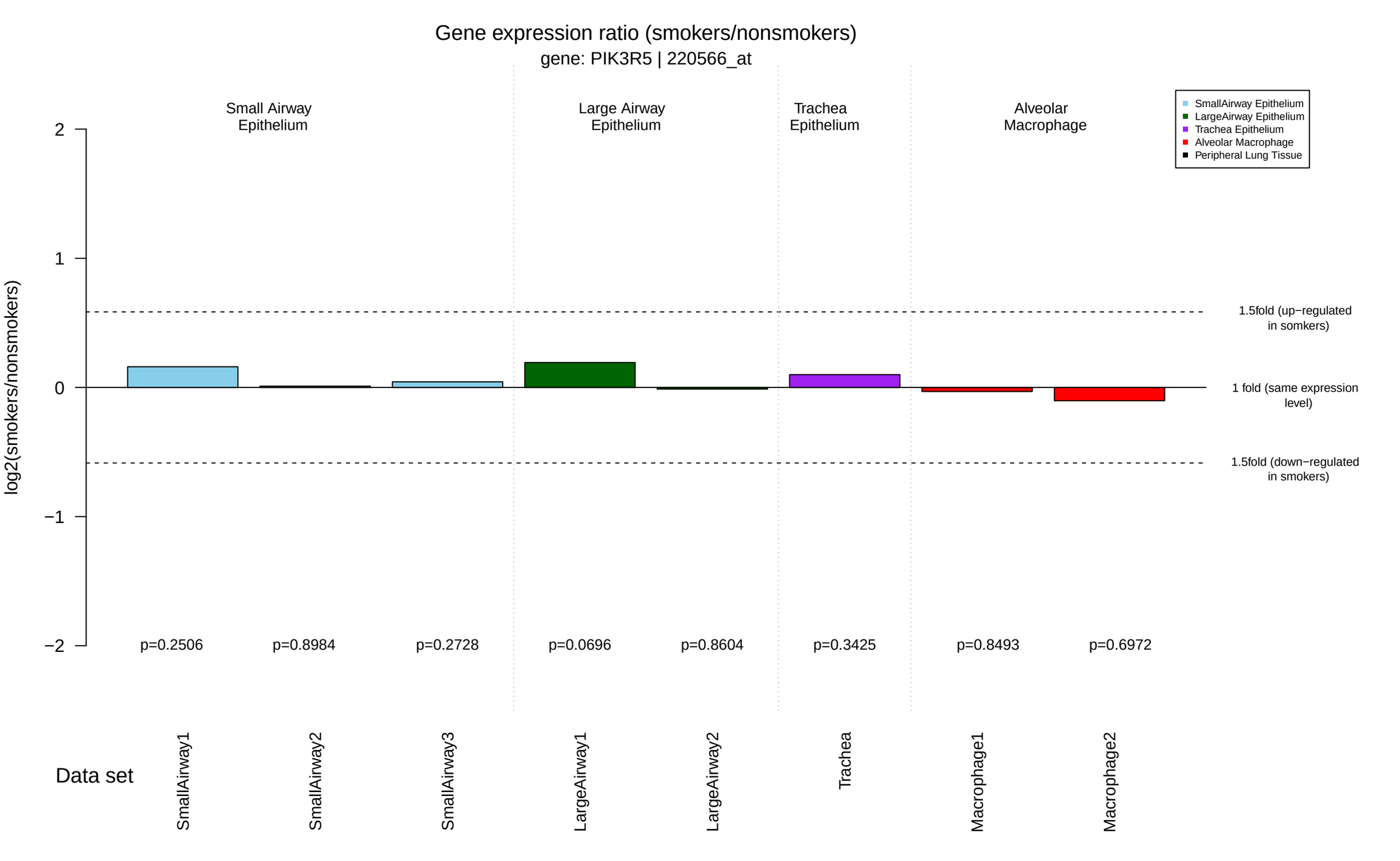
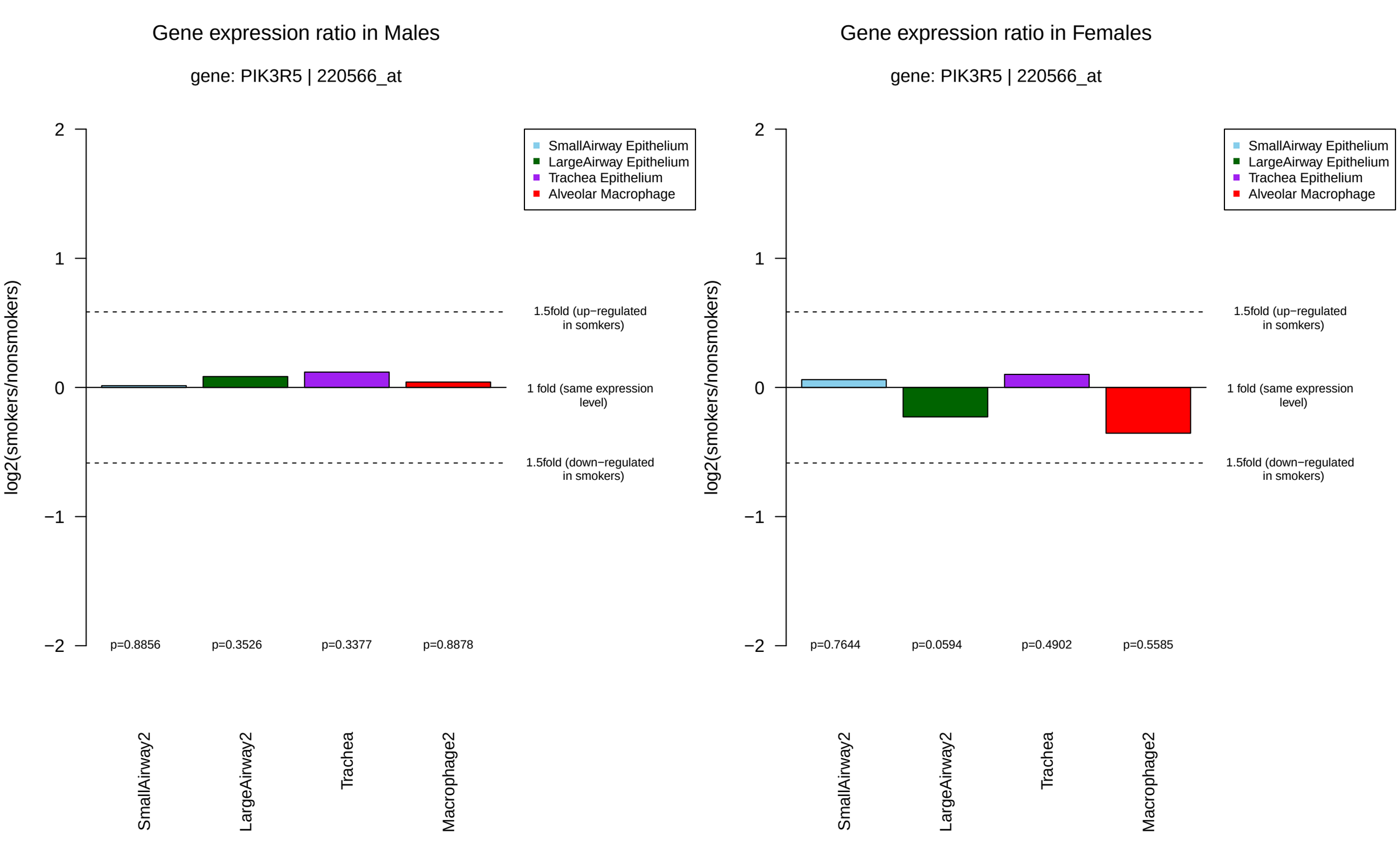
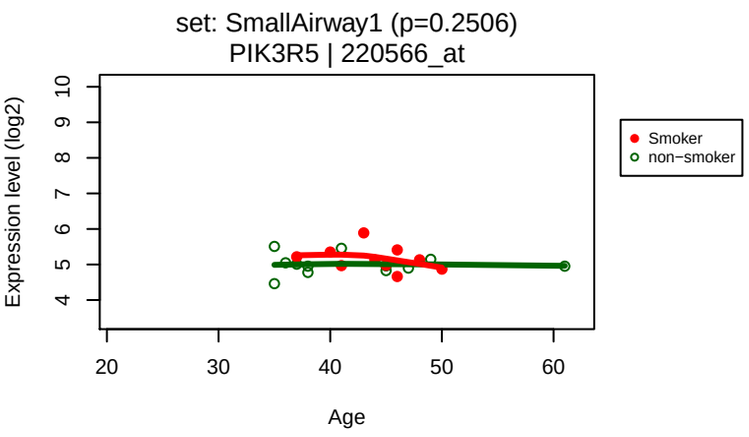
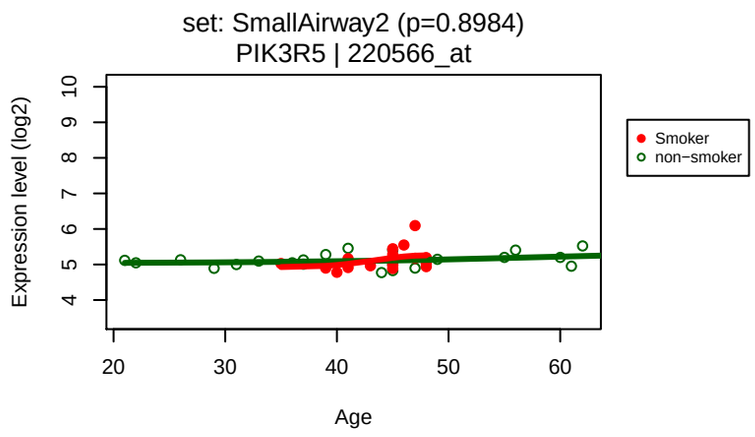
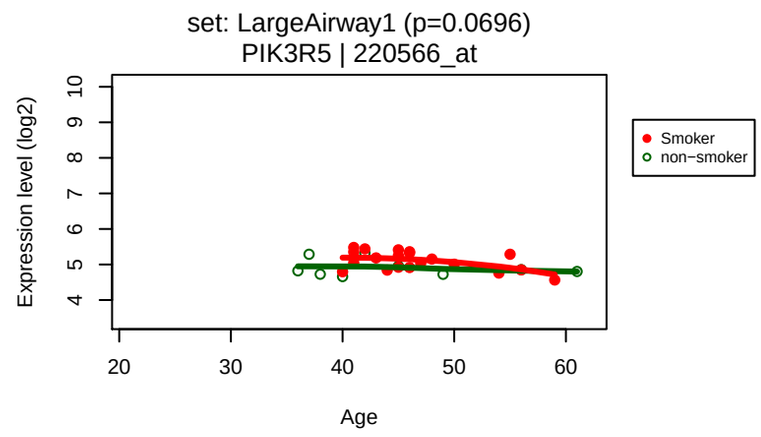

Gene expression of smokers and nonsmokers
gene: PIK3R5 | 220566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2506
p=0.8984
p=0.2728
p=0.0696
p=0.8604
p=0.3425
p=0.8493
p=0.6972
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

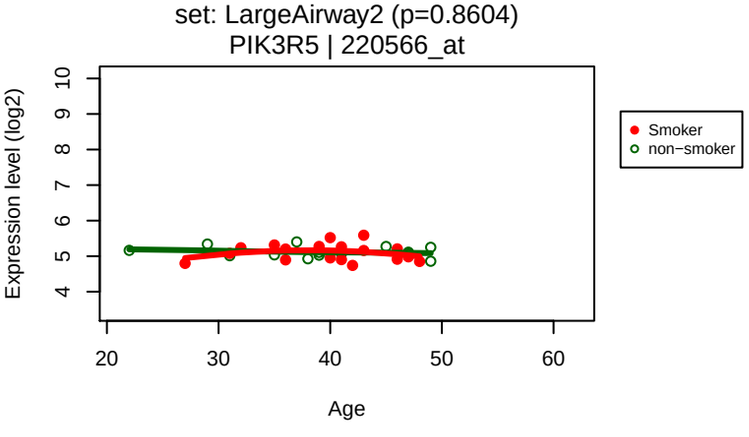

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R5 | 220566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2506
p=0.8984
p=0.2728
p=0.0696
p=0.8604
p=0.3425
p=0.8493
p=0.6972
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

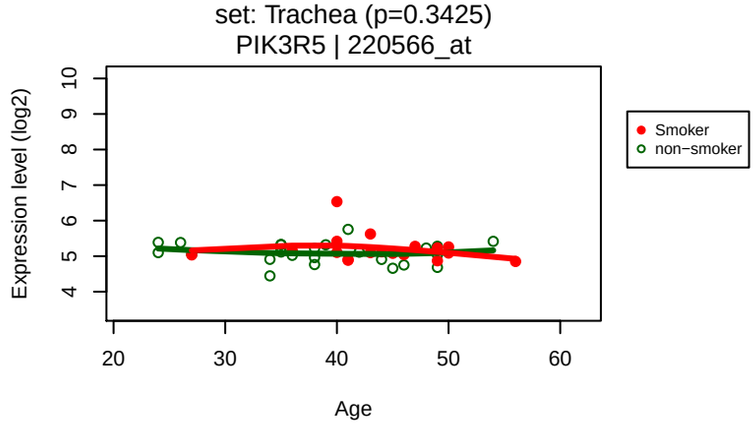

Gene expression ratio in Males
gene: PIK3R5 | 220566_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8856
p=0.3526
p=0.3377
p=0.8878
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R5 | 220566_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7644
p=0.0594
p=0.4902
p=0.5585
SmallAirway2
LargeAirway2
Trachea
Macrophage2

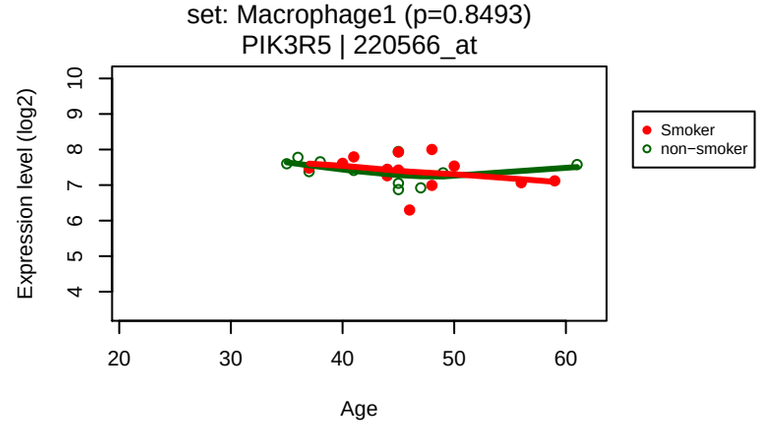

set: SmallAirway1 (p=0.2506)
PIK3R5 | 220566_at
Expression level (log2)
Age
Smoker
non−smoker

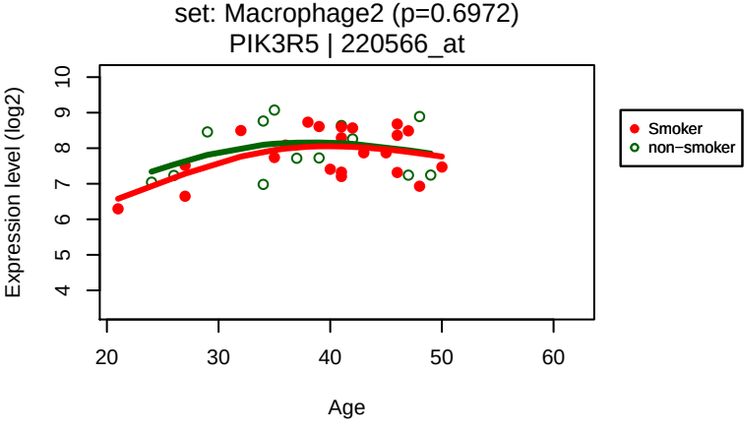

set: SmallAirway2 (p=0.8984)
PIK3R5 | 220566_at
Expression level (log2)
Age
Smoker
non−smoker

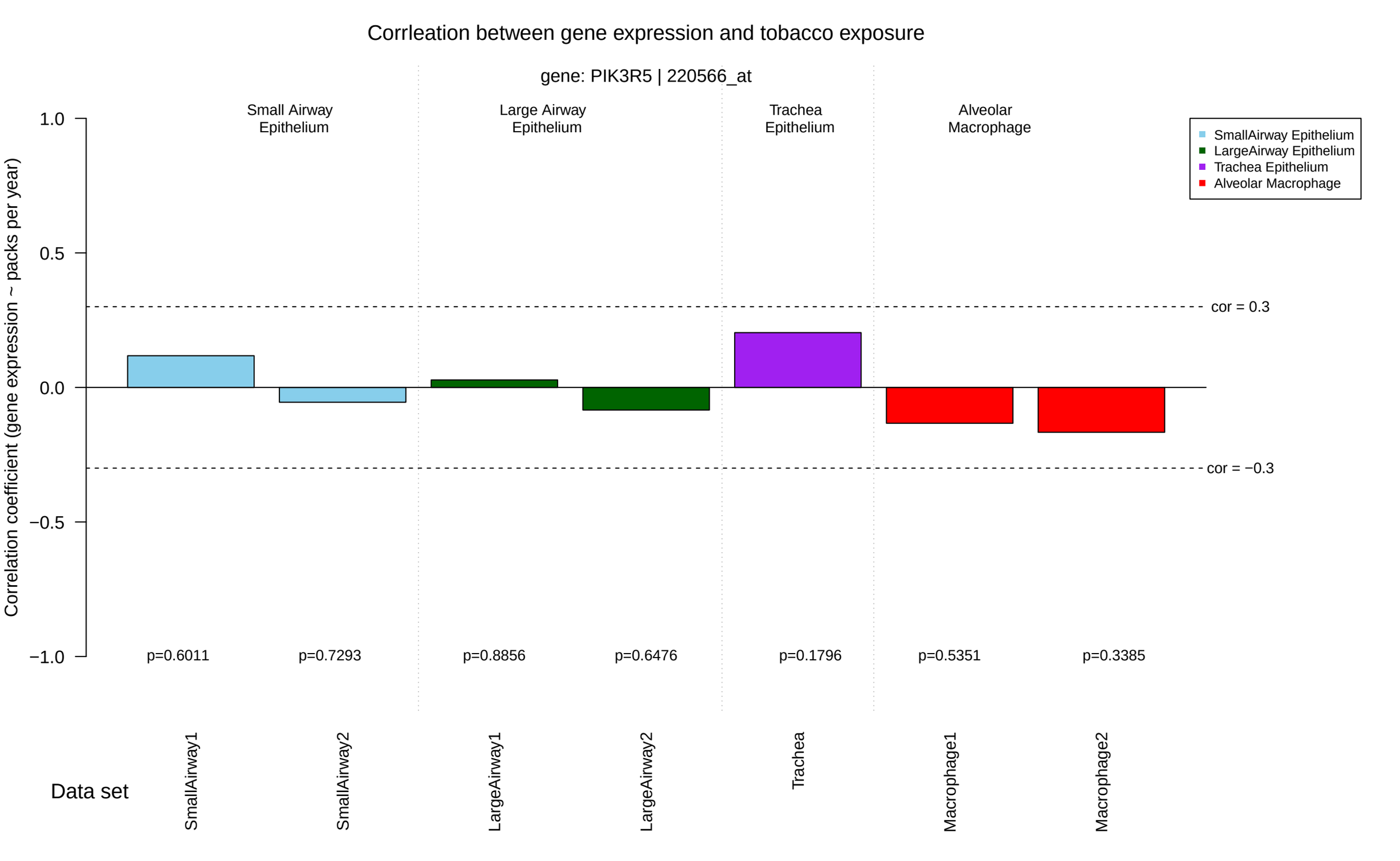

set: LargeAirway1 (p=0.0696)
PIK3R5 | 220566_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8604)
PIK3R5 | 220566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3425)
PIK3R5 | 220566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8493)
PIK3R5 | 220566_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6972)
PIK3R5 | 220566_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R5 | 220566_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6011
p=0.7293
p=0.8856
p=0.6476
p=0.1796
p=0.5351
p=0.3385
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2